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An Adaptive Unsupervised Learning Approach for Credit Card Fraud Detection

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Abstract

Credit card fraud remains a major cause of financial loss around the world. Traditional fraud detection methods that rely on supervised learning often struggle because fraudulent transactions are rare compared to legitimate ones, leading to imbalanced datasets. Additionally, the models must be retrained frequently, as fraud patterns change over time and require new labeled data for retraining. To address these challenges, this paper proposes an ensemble unsupervised learning approach for credit card fraud detection that combines Autoencoders (AEs), Self-Organizing Maps (SOMs), and Restricted Boltzmann Machines (RBMs), integrated with an Adaptive Reconstruction Threshold (ART) mechanism. The ART dynamically adjusts anomaly detection thresholds by leveraging the clustering properties of SOMs, effectively overcoming the limitations of static threshold approaches in machine learning and deep learning models. The proposed models, AE-ASOMs (Autoencoder—Adaptive Self-Organizing Maps) and RBM-ASOMs (Restricted Boltzmann Machines—Adaptive Self-Organizing Maps), were evaluated on the Kaggle Credit Card Fraud Detection and IEEE-CIS datasets. Our AE-ASOM model achieved an accuracy of 0.980 and an F1-score of 0.967, while the RBM-ASOM model achieved an accuracy of 0.975 and an F1-score of 0.955. Compared to models such as One-Class SVM and Isolation Forest, our approach demonstrates higher detection accuracy and significantly reduces false positive rates. In addition to its performance, the model offers considerable computational efficiency with a training time of 200.52 s and memory usage of 3.02 megabytes.

Keywords: credit card fraud detection; unsupervised learning; adaptive reconstruction threshold; self-organizing maps; autoencoder; restricted Boltzmann machines; ensemble learning

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1. Introduction

Credit card fraud presents a critical challenge to the modern financial system, encompassing unauthorized transactions that range from illegal purchases to sophisticated identity theft and card cloning schemes. Global fraud losses reached USD 28.65 billion in 2019 and are projected to exceed USD 43 billion by 2025, with the United States alone reporting USD 11.4 billion in losses in 2020, a 44.7% increase from the previous year [1]. The United Kingdom reported GBP 551.3 million in credit and debit card fraud losses in 2023, with 2.73 million cases averaging GBP 204 per incident, while Japan and Australia

have also experienced considerable losses due to increasingly organized cybercriminal activity. These figures underscore the imperative for developing resilient, scalable fraud detection systems that can dynamically adjust to evolving attack strategies [2].

Financial institutions rely on fraud detection models embedded in transaction monitoring systems to identify fraudulent activities with minimal disruption to legitimate users. These systems prioritize a high F1-score to capture fraudulent transactions while maintaining low false positive rates (FPRs) to reduce operational costs estimated at USD 3.13 per dollar lost to fraud [3]. However, conventional supervised learning methods such as Random Forest, XGBoost, and Support Vector Machines (SVMs) are heavily dependent on large, labeled datasets for training. This presents a significant challenge, as fraudulent transactions typically represent less than 0.2% of all transactions, resulting in severe class imbalance. Models trained on such data are often biased toward the majority class, leading to high false negatives and a failure to detect new fraud patterns [4].

Algorithms such as Autoencoders, Self-Organizing Maps (SOMs), and Restricted Boltzmann Machines (RBMs) are designed to autonomously learn the patterns of normal transactions and flag deviations as potential fraud. This approach is suitable for identifying new, unseen fraud types. Nevertheless, existing unsupervised systems frequently employ a static threshold for anomaly classification. This is a critical limitation, as a single, global threshold fails to account for the inherent complexities and non-uniform distributions within anonymized, high-dimensional financial datasets [4]. Furthermore, most ensemble approaches lack adjustable threshold adaptation informed by clustering capabilities, particularly in PCA-transformed feature spaces used to protect customer privacy [3]. For instance, ref. [5] employed an autoencoder that achieved a 95% detection rate, while ref. [6] achieved low false positives by using clustering techniques. Additionally, a hybrid method integrating unsupervised learning with network analysis improved detection accuracy for complex fraud patterns. However, these approaches often overlook adaptive thresholding prediction and computational efficiency [7].

To this end, this study proposes an ensemble approach that integrates Autoencoders, SOMs, and RBMs with a dynamic Adaptive Reconstruction Threshold (ART) mechanism. Unlike static threshold methods, ART dynamically adjusts decision boundaries using reconstruction error and SOM-based sparse cluster identification. The autoencoders and RBMs are trained exclusively on legitimate transactions to build a robust model of normal behavior. Also, the SOM maps the entire dataset to uncover its underlying topological structure. ART then leverages this structure to define localized anomaly thresholds based on the distribution of reconstruction errors within specific SOM clusters, thereby improving sensitivity to hidden fraud patterns while maintaining low false positive rates.

To further enhance efficiency, we also incorporate Random Forest-based feature selection as a preprocessing step to reduce computational overhead without compromising detection performance [7]. By building on prior work in anomaly detection [5], clustering [7], and dimensionality reduction [3], our approach presents a robust, scalable, and adaptive solution for fraud detection in imbalanced, high-dimensional credit card datasets. This paper offers the following key contributions:

To introduce a novel ART model that dynamically adjusts detection thresholds by integrating SOM clustering on Autoencoders and RBMs, thereby overcoming the limitations of static thresholding.

To provide a comparative analysis of three unsupervised models (Autoencoders, SOMs, RBMs) for credit fraud detection.

To evaluate the models on Kaggle's PCA-transformed dataset and the IEEE-CIS raw feature data, ensuring generalizability across different unseen data.

The remainder of this paper is structured as follows: Section 2 reviews related work and positions our contributions within the existing literature. Section 3 provides a detailed methodology, including the ART algorithm formulation. Section 4 presents comprehensive experimental results and discussion. Section 5 discusses the conclusion and future research directions.

2. Literature Review

The rapid growth of digital transactions has amplified the risk of credit card fraud, necessitating advanced detection methods. This section reviews fraudulent transaction characteristics and recent deep learning approaches.

2.1. Fraudulent Credit Card Transactions

Fraudulent credit card transactions refer to unauthorized activities involving the misuse of credit card information, leading to significant financial losses for individuals, businesses, and financial institutions. These transactions are a growing concern in the digital age, where there is increasing reliance on online and digital payments, as it is an avenue for fraudsters to exploit vulnerabilities in payment systems. One common method of fraud is skimming, where malicious devices are placed on automated teller machines (ATMs) or point-of-sale (POS) terminals to capture card data during legitimate transactions. The stolen information is then used to produce counterfeit cards or carry out unauthorized purchases [4]. Account takeover is another form of credit card fraud in which attackers gain unauthorized access to a victim's existing account, often by exploiting weak passwords or leveraging data obtained through security breaches. Once access is obtained, the fraudster may modify account details and carry out unauthorized transactions [5].

Detecting credit card fraud presents distinct challenges that set it apart from general anomaly detection tasks [8,9]. The key difficulty lies in the extreme class imbalance, which creates a scenario that conventional classification methods often fail to handle effectively [10]. Additionally, the high cost associated with false positives further complicates detection efforts [7].

2.2. Deep Learning Algorithms for Credit Card Fraud Detection

Deep learning models, such as Recurrent Neural Networks (RNNs) and Convolutional Neural Networks (CNNs), have proven effective in capturing complex temporal and spatial patterns. Long Short-Term Memory (LSTM) networks, a variant of RNNs, are particularly adept at identifying sequential patterns within transaction histories. Meanwhile, CNNs can be adapted for fraud detection to analyze spatial relationships between transaction attributes [11]. Also, Feedforward Neural Networks (FNNs) serve as foundational models for fraud detection, leveraging their ability to learn relationships between transaction features to distinguish between legitimate and fraudulent activities [12]. However, these methods remain dependent on labeled data and suffer from the fundamental class imbalance problem [13].

Supervised learning methods have dominated fraud detection research due to their strong performance when sufficient labeled data is available. Random Forest, XGBoost, and Gaussian Mixture Models (GMMs) have accurately distinguished fraudulent from legitimate transactions. However, Random Forest faces scalability issues in large-scale credit risk assessment, and XGBoost struggles with highly imbalanced real-world datasets [14,15]. Also, GMM shows reduced performance on skewed data distributions [16]. In contrast, unsupervised methods like Isolation Forest, Generative Adversarial Networks [17,18], and K-Means Clustering [19] enable the detection of fraudulent patterns without labeled data, offering enhanced scalability and adaptability to evolving fraud schemes.

The advantages of unsupervised learning in fraud detection are numerous. These methods exhibit adaptability by effectively detecting emerging fraud patterns without needing labeled data. They are also scalable and capable of processing large datasets efficiently without the overhead of manual labeling. Furthermore, unsupervised techniques are less prone to bias associated with imbalanced datasets, as they focus on identifying anomalies rather than relying on predefined labels [20,21]. This is especially valuable in credit card fraud detection, where fraudulent transactions are far outnumbered by legitimate ones. By tackling the drawbacks of supervised methods, such as class imbalance, unsupervised learning models like Autoencoders, Self-Organizing Maps (SOMs), and Restricted Boltzmann Machines (RBMs) play a crucial role in modern fraud detection systems.

2.3. Related Work

Traditional supervised machine learning approaches often struggle with identifying novel fraud patterns, which has led to an increasing interest in unsupervised learning techniques for fraud detection.

Unsupervised learning presents a compelling solution to the inherent limitations of supervised methods in credit card fraud detection. In 2023, authors in [17] proposed UAAD-FDNet, a GAN-based model that achieved an F1-score of 0.85 on the Kaggle and IEEE-CIS datasets. Despite this, the model exhibited high false positive rates, limiting its real-world applicability. Several ensemble approaches have been used in studies to improve detection by combining multiple models. However, common limitations of high false positives and low recall persist. Many existing methods still use fixed thresholds, a simple training method that may not utilize the strengths of individual models, and does not make good use of clustering to improve threshold settings. Although quantum kernel-based One-Class SVMs have shown better precision and F1-score, they are limited by high computational demands [22]. Similarly, optimization-based models such as Bat Optimization Algorithm combined with CNNs offered strong accuracy but lacked scalability for deployment [23].

Graph-based methods have been widely applied to provide contextual insights for anomaly detection. For example, Sergadeeva et al. [23] used Graph Neural Networks to model transaction relationships, enhancing anomaly detection but at a high computational cost. Hewapathirana [24] employed OLS regression, Random Forest, and Dropout Neural Networks to detect outliers based on transaction deviations. Although promising for real-time detection, DNNs produced many false positives. Similarly, SOM-based approaches have shown strong performance. Other studies explored One-Class SVMs and autoencoders, noting issues with precision and recall. Comparative work by Hancock et al. [25] showed that while Logistic Regression performed well, it failed to model complex fraud patterns as effectively as CNNs.

Autoencoders have been used to reconstruct normal transactions and flag high-reconstruction-error cases as fraud. Anurag MitraMukul [26] and Rezapour [27] all found Autoencoders effective in highly imbalanced datasets, though low precision and recall remained a concern. Incorporating techniques like SMOTE or advanced feature selection could improve performance.

Hariteja [28] compared Isolation Forest, Local Outlier Factor, and One-Class SVM on the Kaggle dataset. Isolation Forest achieved 99.74% accuracy, outperforming others. However, the study lacked key performance metrics like F1-score and was limited to a single dataset, raising concerns about generalizability. Prediction-interval-based models have also emerged. Ileberi and Sun [29] treated fraud detection as a clustering task, mapping normal and anomalous transactions to different feature space regions. Their

method outperformed traditional models on the Kaggle dataset but faced scalability issues with high-dimensional data.

Despite recent advances, several key limitations persist. A primary challenge is the reliance of many models on static thresholds, which are often unsuited for the dynamic characteristics of high-dimensional, anonymized datasets [30,31]. Moreover, ensemble methods frequently lack the adjustment mechanisms required to leverage clustering for threshold adaptation. High computational demands also pose a significant challenge. The limited validation across diverse datasets further restricts the generalizability of these approaches.

In response to these challenges, we propose the Adaptive Reconstruction Threshold (ART) method. This approach integrates Self-Organizing Map (SOM) clustering to enable dynamic threshold adaptation. By leveraging the topological properties of the SOM, ART refines reconstruction error boundaries, a key distinction from traditional ensemble models. This process not only enhances model performance but also maintains computational efficiency. Table 1 summarizes existing techniques for credit card fraud detection, positioning our work within the broader literature.

Table 1. Comparison of some existing techniques used in credit card fraud detection.

Authors	Algorithm Used	Dataset	Accuracy	F1-Score
Jiang et al. [17]	GAN | Autoencoder	Kaggle Credit Card, IEEE-CIS Fraud Detection	×	85.2
Kyriienko and Magnusson [22]	Quantum Kernel OC | SVM	Simulated Dataset (20 Qubits)	×	15
Sergadeeva et al. [23]	BOA | CNN | FCL	Kaggle Credit Card Dataset	99.0	90.0
Hewapathirana [24]	OLS, RF, DNN	Real-World Transaction Data	94.8	2.2
Hancock et al. [25]	Logistic Regression, K-Means, CNN	Kaggle Credit Card Dataset	×	90.5
MitraMukul et al. [26]	Autoencoder	European Credit Card Dataset	97.0	11.0
Rezapour et al. [27]	Autoencoder | One-Class SVM	Real-World Credit Card	95.0	83.0
Hariteja [28]	Isolation Forest, LOF, One-Class SVM	Kaggle Credit Card Dataset	99.7	×
Iieberi et al. [29]	CNN, LSTM, XGBoost	European Credit Card Dataset	96.0	94.1
Our Models	Autoencoder, SOM, Restricted Boltzmann Machines	Kaggle, IEEE-CIS Fraud Detection	See Section 4	See Section 4

3. Methods

This section details the experimental setup, datasets, and the proposed ensemble unsupervised model, which integrates the Adaptive Reconstruction Threshold (ART) approach. The analysis was performed on a Windows 11 Enterprise system equipped with an Intel® Core™ i7-10700 CPU (2.90 GHz), 16 GB RAM, and an NVIDIA Quadro P620 GPU. The implementation was developed in Python 3.9.1 using libraries such as TensorFlow 2.3.0, Keras 2.7.0, Scikit-learn 1.0.2, and MiniSom 2.2.9.

The proposed fraud detection workflow, as shown in Figure 1, is designed to evaluate a novel adaptive thresholding mechanism employing two distinct datasets: the Kaggle credit card fraud dataset [32] and the IEEE-CIS fraud detection dataset [33] Two models, an Autoencoder and a Restricted Boltzmann Machines (RBM), are trained using the Kaggle data. Next, a Self-Organizing Map (SOM) is applied to the full Kaggle dataset to visualize transaction patterns and identify sparse clusters. This process adjusts the initial cutoff into a more adaptive threshold. The updated threshold is evaluated on both the Kaggle and IEEE-CIS test sets. Model performance was evaluated using a suite of standard metrics, including accuracy, precision, recall, and the Area Under the Receiver Operating Characteristic Curve (AUC), F1-score, false positive rate (FPR), and false negative rate (FNR). These results are compared against a basic static threshold $(\mu + 2\sigma)$, to provide a complete performance analysis and validate the model's robustness across different datasets. Additionally, the computational resources required by each model were assessed to evaluate efficiency.

The architecture of the proposed models follows a structured approach, as shown in Figure 2. Autoencoders and Restricted Boltzmann Machines (RBMs) are trained on normal transactions to model legitimate behavior, using reconstruction error and divergence metrics to detect anomalies. On the other hand, Self-Organizing Maps (SOMs) are trained on normal and fraudulent transactions, clustering them on a grid to identify outliers. Anomaly detection classifies transactions as normal or fraudulent, followed by a model evaluation to assess performance. Hyperparameter tuning is iteratively applied before and after evaluation to optimize each model until performance criteria are met. This is followed by evaluation metrics and visualization, where metrics such as construction and quantization errors are used in Dataset 1. Finally, the process results in a binary classification, labeling each transaction as either fraudulent or non-fraudulent.

3.1. Dataset

The Kaggle Credit Card Fraud Detection dataset [32] contains 284,807 PCA-transformed transactions with 28 principal components (V1–V28) plus 'Time' and 'Amount' features. The dataset exhibits severe class imbalance, with only 492 fraudulent transactions (0.172% fraud rate). Meanwhile, the IEEE-CIS Fraud Detection dataset [33] was utilized for validation. The dataset, originally provided in four files (train transaction, train identity, test transaction, and test identity), was concatenated into a single dataframe using TransactionID as the primary key [33]. This initial merging resulted in a dataset of 590,540 transaction samples and 433 features.

Subsequently, features were removed based on two criteria: relevance and data completeness. Some features, such as transaction date, were dropped, and 378 features containing a substantial amount of missing data were excluded to prevent them from adversely affecting model performance. Similar to the Kaggle dataset, this data also shows a significant class imbalance, with fraudulent transactions comprising approximately 3.5% of the total training data.

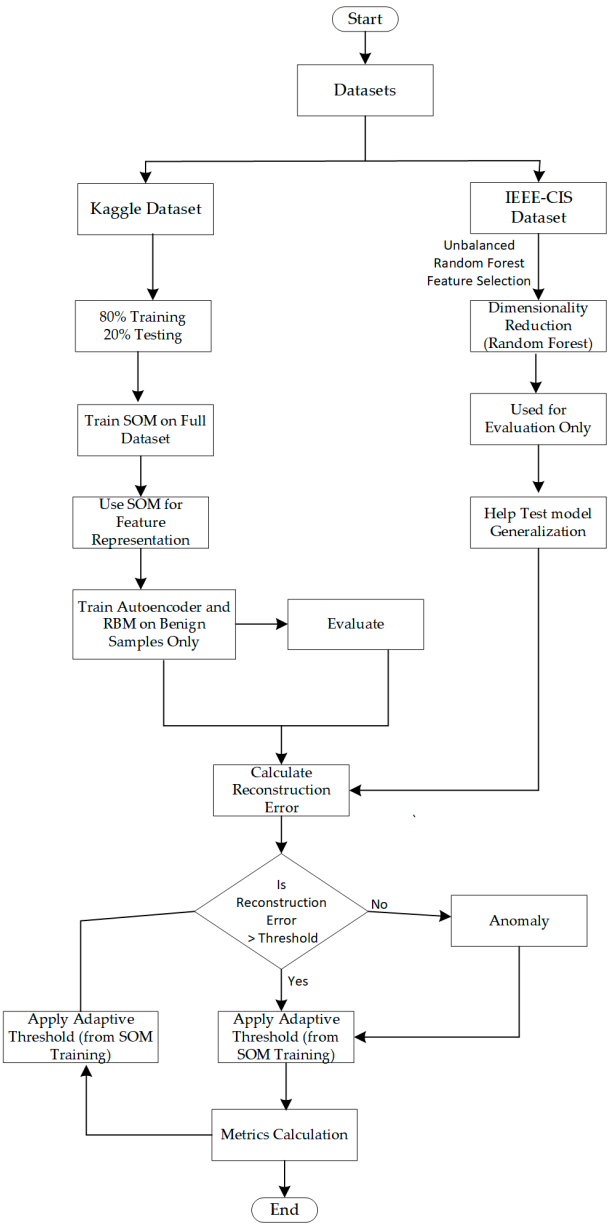


Figure 1. Flowchart of the proposed method.

Adhering to the unsupervised learning paradigm for anomaly detection, the Autoencoder and Restricted Boltzmann Machines (RBM) were trained exclusively on normal transactions from the training set. Alternatively, the Self-Organizing Map (SOM) was trained on the complete Kaggle dataset, including both normal and fraudulent transactions, to capture a comprehensive representation of its patterns. Both the validation and testing sets contained a mix of fraudulent and non-fraudulent transactions to enable a proper performance assessment. The IEEE-CIS Fraud Detection dataset was reserved for cross-validation, serving as a crucial test set to ensure the model's generalizability across different data distributions.

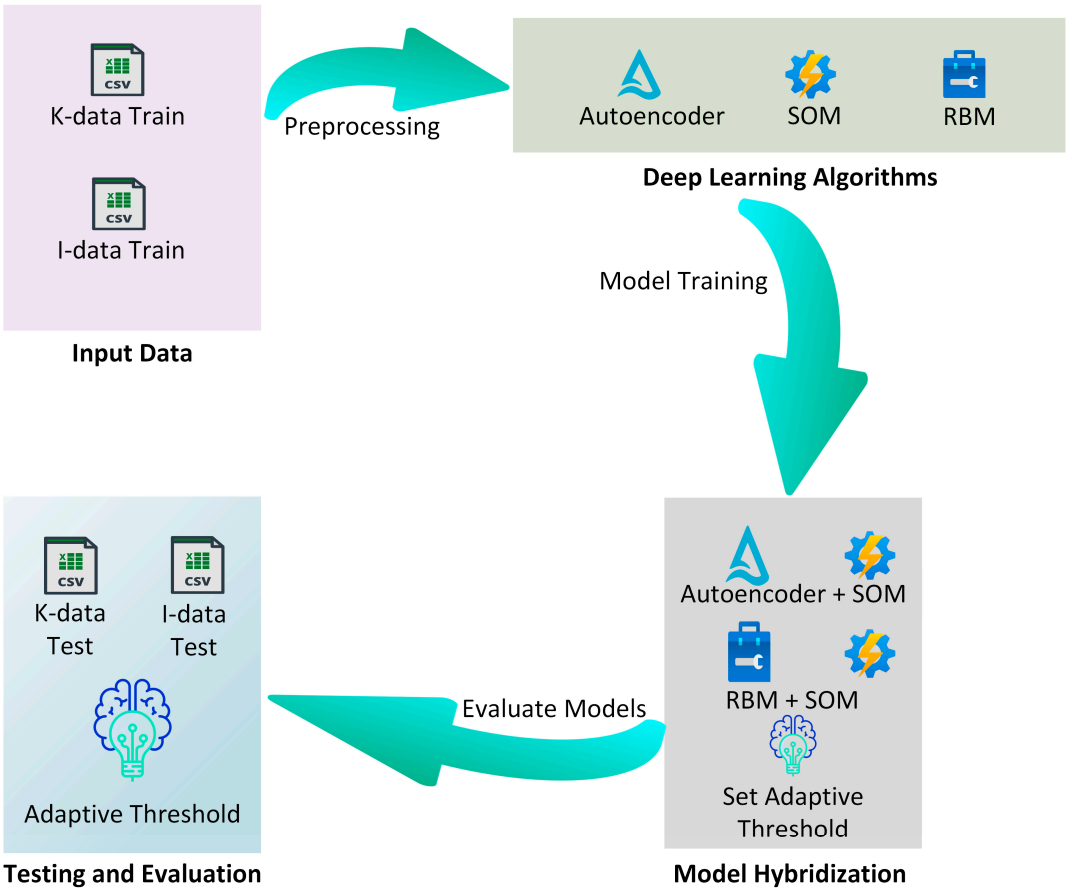


Figure 2. The architecture of the proposed model.

3.2. Preprocessing

Extensive preprocessing was performed on the datasets to ensure accurate and reliable model training. Missing values were handled by removing transactions with missing amounts and employing median imputation for other features. All numerical features were normalized using Min-Max scaling [34] a transformation applied with the following formula:

$$X' = \frac{X - X_{min}}{X_{max} - X_{min}} \quad (1)$$

Categorical variables were converted using one-hot encoding [35]. For dimensionality reduction, Principal Component Analysis (PCA) was performed on the IEEE-CIS dataset, reducing it from 433 to 20 features while retaining 95% of the original variance. The existing PCA transformation of the Kaggle dataset was preserved to maintain its anonymization properties.

We adopted a training method that reflects real-world fraud detection scenarios to maintain data integrity and avoid bias introduced by resampling methods. Specifically, the Autoencoders and Restricted Boltzmann Machines (RBMs) were trained solely on normal transactions, enabling the models to learn patterns of legitimate behavior. The SOM model's clusters were analyzed to determine which areas corresponded to anomalies, enabling a visual and data-driven interpretation of abnormal behavior.

3.3. The Proposed Models

The Adaptive Reconstruction Threshold (ART) method dynamically adjusts detection thresholds using SOM clustering, enhancing accuracy in high-dimensional, PCA-transformed datasets. These models were chosen over alternatives like Generative Adver-

sarial Networks (GANs), which incur high computational costs, or Isolation Forest, which lacks adaptability to dynamic fraud patterns, ensuring a balance of precision, scalability, and efficiency [35].

3.3.1. Autoencoder

An Autoencoder [36] is a neural network designed to learn a compressed, latent representation of input data. It achieves this by minimizing the reconstruction error between the network's output and the original input, $x \in R^n$ and its reconstruction X. The Autoencoder consists of two components:

- The **encoder** function $f(x;\ \theta)$ maps the input to a lower-dimensional latent space $z\ =\ f(x;\ \theta)$,
- The **decoder** function $g(z; \varphi)$ maps the latent space back to the original input space $\overline{X} = g(z; \varphi)$

The Autoencoder is trained to minimize the Mean Squared Error (MSE) loss on normal transactions. This loss function, which quantifies the reconstruction error between the input and output, is defined as [36]

$$L_{MSE} = \frac{1}{n}\sum_{i=1}^{n} \parallel X_i - \overline{X} \parallel^2 \tag{2}$$

where $||.||_2$ denotes the Euclidean norm, and n represents the number of data points. In this paper, the Autoencoder is trained exclusively on normal transactions from the Kaggle Credit Card Fraud Detection dataset, enabling it to learn patterns characteristic of typical behavior. Anomalies are then detected by comparing the reconstruction error of test transactions to a predetermined threshold; transactions that exceed this threshold are classified as fraudulent. The Autoencoder was chosen for its effectiveness in learning compact representations of high-dimensional data and its ability to reconstruct inputs based on these learned patterns.

3.3.2. Self-Organizing Maps (SOMs)

The Self-Organizing Map (SOM) [37] is an unsupervised learning model that projects high-dimensional data onto a lower-dimensional grid while preserving the structure of the input space. It consists of a lattice of neurons, each associated with a weight vector. $W_j \in R^n$, where n is the dimensionality of the input data. In this study, the SOM projects transaction data onto a 10×10 grid, and its weights are updated through a process of competitive learning. This process enables the clustering of both normal and fraudulent transactions, with sparse clusters indicating potential anomalies. For a given input vector x, the Best Matching Unit (BMU) is the neuron whose weight vector is closest in Euclidean distance to the input. The BMU is identified as the neuron c that minimizes the following:

$$j* = \text{argmin} n_j \left\| x - w_j \right\| \tag{3}$$

During training, the BMU's weights and its neighboring neurons are updated to approach x according to the learning rule:

$$w_j(t+1) = w_j(t) + \eta(t) h_{j,j_*}(t)(x - w_j(t)) \tag{4}$$

The learning rate, $\eta(t)$, and the neighborhood function, $h_{j,j_*}(t)$, govern the update magnitude of the SOM's weights based on the distance between neuron j and the Best Matching Unit (BMU), $j*$. We trained the SOM on the full Kaggle Credit Card Fraud Detection dataset (including both normal and fraudulent transactions). This approach allowed

the SOM to cluster transactions based on shared features, capturing a comprehensive representation of both normal and anomalous patterns. The SOM was selected for its powerful unsupervised clustering capabilities and its ability to identify distinct transaction clusters without the need for explicit class balancing, a key advantage over supervised methods.

The SOM serves two purposes in our methodology. First, it performs independent anomaly detection by measuring the quantization error of test transactions from their BMUs, with outliers identified when this error exceeds a static threshold. Second, it supports the Adaptive Reconstruction Threshold (ART) for the Autoencoder and RBM by identifying sparse clusters within the Kaggle dataset that correspond to fraudulent transactions. The SOM provides important information to adjust the reconstruction error thresholds dynamically.

3.3.3. Restricted Boltzmann Machines (RBM)

The Restricted Boltzmann Machines (RBM) [38] is a generative neural network model comprising visible units v (representing the input) and hidden units h, with no intra-layer connections. RBM models normal transaction distributions using energy functions, trained with Contrastive Divergence, 256 visible, 128 hidden units. It defines a joint probability distribution over the visible and hidden units, such as

$$P(v,h) = \frac{1}{z} exp(-E(v,h)) \tag{5}$$

where z is the partition function, and $E(v,h)$ is the energy function defined as

$$E(v,h) = -v^T Wh - b^T v - c^T h \tag{6}$$

Here, W is the weight matrix connecting visible and hidden units, while b and c are the biases of the visible and hidden layers, respectively. The RBM is trained to minimize the energy for the visible layer corresponding to normal transactions. Like the Autoencoder, the RBM was trained exclusively on normal data to establish a baseline for non-fraudulent behavior. Anomalies were subsequently identified by their high reconstruction error, which corresponds to a higher energy level and indicates a poor fit to the learned distribution. The RBM was chosen for its probabilistic nature and its effectiveness in modeling complex data distributions, which allows it to capture the statistical structure of normal transactions.

3.3.4. Adaptive Reconstruction Threshold for Anomaly Detection

In reconstruction-based anomaly detection models such as Autoencoders and Restricted Boltzmann Machines (RBMs), selecting an appropriate threshold to differentiate between normal and fraudulent transactions is a critical challenge. This is particularly true when working with datasets transformed by Principal Component Analysis (PCA), where features are anonymized. To address this, we introduce the Adaptive Reconstruction Threshold (ART) method, which dynamically adjusts thresholds by integrating statistical error analysis with Self-Organizing Map (SOM) clustering.

The ART method operates in two stages. First, a preliminary threshold T_0 is established by setting it to the 95th percentile of reconstruction errors $E = \{e_i\}$ from normal transactions in the training set. This provides a robust initial threshold that effectively captures significant reconstruction errors while limiting false positives. Second, ART adjusts this threshold to T from the SOM. The method identifies sparse clusters defined as

those containing less than 5% of total mappings, and adjusts the threshold to maximize the detection of anomalies within these clusters. This optimization is formalized by [4]

$$T = \text{argmax}_t \sum_{x_i \in C_s} \frac{\text{II}(e_i > t)}{|C_s|} \quad (7)$$

where the probability of a transaction being an anomaly, given its location in a sparse cluster, is at least 0.8. This dynamic adjustment overcomes the limitations of static thresholds ($\mu + 2\sigma$) and ensures high sensitivity to hidden anomalies that often reside in the sparse regions of the feature space [35,38]. The ART method relies exclusively on training data for threshold adjustment, thereby preventing test data leakage [39,40].

3.3.5. The Model Implementation

The models were trained on the Kaggle Credit Card Fraud Detection dataset and evaluated using the IEEE-CIS Fraud Detection dataset. The Autoencoder was implemented in TensorFlow, consisting of three hidden layers in both the encoder and decoder, with layer sizes of 64, 32, and 16 units, respectively. Rectified Linear Unit (ReLU) activation functions were employed to capture nonlinear patterns in the data. The model was trained exclusively on normal transactions to minimize reconstruction error using the Mean Squared Error (MSE) loss function. Hyperparameters, including learning rate, batch size, and number of epochs, were optimized using a grid search approach. The best-performing configuration, identified by monitoring performance on a validation set, was a learning rate of 0.001, a batch size of 64, and 50 training epochs.

The Self-Organizing Map (SOM) was employed to cluster transactions and detect outliers. Implemented using the MiniSom library, it organized normalized transaction features onto a two-dimensional 10×10 grid, where each node represented a cluster of similar transactions. Unlike the Autoencoder, the SOM [41] was trained on the complete dataset (both normal and fraudulent transactions), enabling it to learn a comprehensive topological structure that distinguished the two classes. Training was conducted via a competitive learning process. Key hyperparameters, including grid size, learning rate, and the number of training iterations, were tuned. The final configuration utilized a 10×10 grid, a learning rate of 0.001, and 500 iterations.

The Restricted Boltzmann Machines (RBM), implemented using the TensorFlow Probability library, was a generative model designed to learn the underlying probability distribution of normal transactions. The RBM consisted of 256 visible units, to match the input features, and 128 hidden units. Like the Autoencoder, the RBM was trained exclusively on normal transactions to avoid bias toward the majority class. Training was performed using the Contrastive Divergence algorithm to minimize the model's energy function. Hyperparameters were optimized using grid search, resulting in an optimal configuration that comprised a learning rate of 0.001, a batch size of 64, and five Gibbs steps.

The parameters for the Adaptive Reconstruction Threshold (ART) were also optimized by monitoring validation loss. The preliminary threshold was set at the 95th percentile, and the sparse cluster threshold for the SOM was set below 5% mappings. These parameters were iteratively adjusted to maximize the F1-score, yielding an optimal model configuration that balanced computational efficiency with anomaly detection accuracy. Table 2 further shows the hyperparameter configuration break down after it was validated on Dataset 1, and a conclusive evaluation was performed on Dataset 2.

Table 2. A summary of the different parameter settings for the models.

Algorithms	Parameters Used	Values
Autoencoder	Input_dim	28
	Latent_dim	16
	Hidden_layers	64, 32, 16 (encoder/Decoder)
	Learning rate	0.001
	Batch_size	64
	Epochs	50
	Activation	ReLU
	Loss_function	MSE
	Optimizer	Adam
SOM	Grid_size	10×10
	Learning_rate	0.5
	Iterations	500
	Activation	Competitive learning
RBM	Visible_units	256
	Hidden_units	128
	Learning_rate	0.001
	Batch_size	64
	Gibbs_steps	5
	Training_algorithm	Contrastive Divergence (CD-1)

Algorithms 1 and 2 detail the proposed Adaptive Reconstruction Threshold (ART)-based ensemble model, which integrates an Autoencoder, a Self-Organizing Map (SOM), and a Restricted Boltzmann Machines (RBM) with an initial Random Forest feature selection. The models are trained exclusively on normal transactions from the Kaggle dataset and subsequently evaluated on both the Kaggle and IEEE-CIS test datasets. Model performance was evaluated using a rage of metrics, including precision, recall, F1-score, and the Area Under the Receiver Operating Characteristic Curve (AUC-ROC).

Algorithm 1.	Adaptive Reconstruction Threshold (ART) Method
Input:	Training data D_train, Validation data D_val, Test data D_test
Output:	Optimized fraud detection threshold θ*
1.	Extract normal transactions: D_normal_train ← {x ∈ D_train ∣ label(x) = 0}
2.	Initialise autoencoder AE, train on D_normal_train
3.	Compute reconstruction errors: E_train ← {$\|\|x - AE(x)\|\|_2$ ∣ x ∈ D_normal_train}
4.	Set initial threshold: $\theta_0 \leftarrow$ percentile(E_train, 95)
5.	Initialize SOM with appropriate dimensions
6.	som.train(normalize(D_normal_train))//Only normal training data
7.	**for** each validation batch D_val_k in D_val **do**:
8.	Apply current threshold θ_k to get predictions P_k
9.	Compute validation metrics: precision_k, recall_k, F1_k
10.	if F1_k > F1_threshold AND recall_k > 0.8 then:
11.	$\theta_{k+1} \leftarrow$ refine_threshold(θ_k, P_k, D_val_k)//See Algorithm 2
12.	else:
13.	$\theta_{k+1} \leftarrow \theta_k$
14.	return θ* ← θ_finalR_i ← metrics (P_i, test_i_labels)
15.	**end**

Algorithm 2.	Threshold Refinement Process
Input:	Current threshold θ_k, Predictions P_k, Validation batch D_val_k
Output:	Refined threshold θ_{k+1}
1.	Compute false positive rate: FPR_k = FP/(FP + TN)
2.	Compute false negative rate: FNR_k = FN/(FN + TP)
3.	Define adjustment factor: $\alpha = 0.1$
4.	if FPR_k > 0.05 then://Too many false positives
5.	$\theta_{k+1} \leftarrow \theta_k \times (1 + \alpha)$//Increase threshold
6.	else if FNR_k > 0.2 then://Too many false negatives
7.	$\theta_{k+1} \leftarrow \theta_k \times (1 - \alpha)$//Decrease threshold
8.	else:
9.	$\theta_{k+1} \leftarrow \theta_k$//No adjustment needed
10.	return θ_{k+1}
11.	**end**

4. Results and Discussion

This section presents the findings from implementing three unsupervised learning models for credit card fraud detection: the Autoencoder, the Self-Organizing Map (SOM), and the Restricted Boltzmann Machines (RBM). The analysis focuses on key performance and efficiency metrics, including reconstruction/quantization error, loss, training and testing time, and memory utilization. A comprehensive evaluation of each model is provided using standard metrics such as accuracy, precision, recall, F1-score, and the Area Under the Receiver Operating Characteristic Curve (AUC-ROC).

4.1. Training Results of Autoencoder, SOMs, and RBM Models

The evaluation of the proposed Adaptive Reconstruction Threshold (ART) method was conducted using a range of metrics, including accuracy, precision, recall, F1-score, and false positive rate (FPR). These metrics were chosen to align with the operational requirements of financial institutions, which seek to balance detection rates with minimizing false alarms. To ensure robust generalization, five-fold cross-validation was applied to the training set of normal transactions for both hyperparameter tuning and performance validation. The computational complexity of AE-ASOM is $O(n.d.h)$ for the Autoencoder (n transactions, d features, h hidden units), $O(n.m^2)$ for SOM (m grid size), and $O(n.v.h)$ for RBM (v visible units). ART adds $O(n.m^2)$. Figure 3 demonstrates the training progression of all three models over 50 epochs. The Autoencoder achieves stable convergence with training and validation MSE decreasing from 0.07 to 0.01, corresponding to an accuracy improvement from 90% to 98%. The SOM displays rapid convergence within the first 100 iterations, with quantization error stabilizing near zero and clustering accuracy reaching 96%. The RBM shows consistent energy minimization, achieving 97% reconstruction accuracy after 50 epochs.

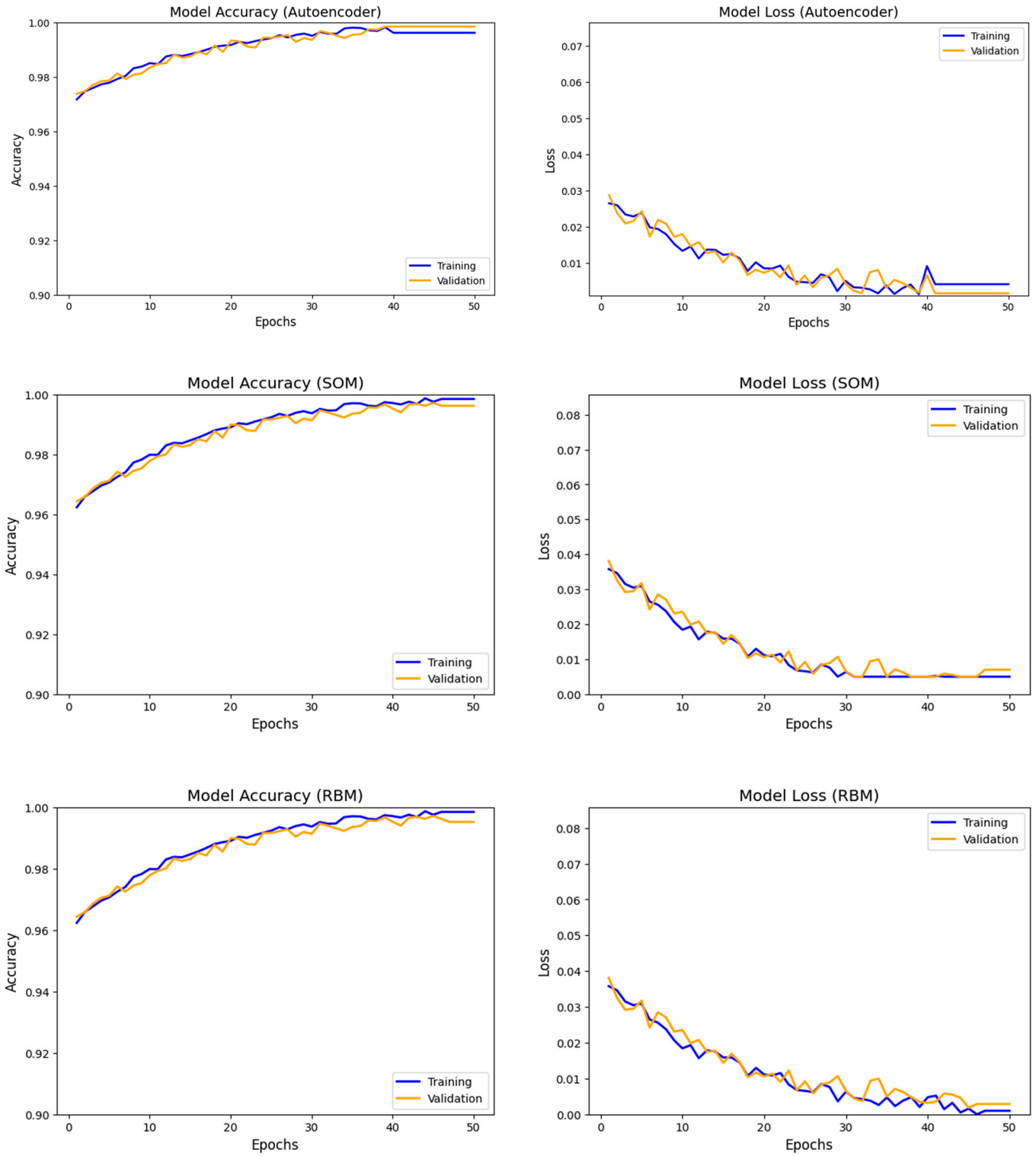


Figure 3. Accuracy and validation loss of the Autoencoder, SOMs, and RBM models.

4.2. Performance Evaluation

The evaluation of the models focused on their effectiveness in identifying anomalies based on reconstruction and quantization errors. Normal transactions consistently yielded low error values, typically remaining below the established thresholds. In contrast, fraudulent transactions produced significantly higher errors, often exceeding these thresholds. This discrepancy is a direct result of the models' training methodologies. The Autoencoder and Restricted Boltzmann Machines (RBM) were trained exclusively on

normal transactions, making them ill-equipped to reconstruct fraudulent ones. Consequently, these anomalous patterns generated higher reconstruction errors. Similarly, the Self-Organizing Map (SOM) assigned fraudulent transactions to sparse or less populated regions of the map, highlighting their deviation from the learned topological structure of normal transaction clusters.

Given the dataset's imbalanced nature, fraud comprised only 0.172% in the Kaggle dataset [32] and 3.5% in the IEEE-CIS dataset [33] models, focused on learning normal behavior, making fraud more detectable as outliers. Visualization of the Self-Organizing Maps grid further illustrated this separation. Non-fraudulent transactions were clustered centrally and to the left, while fraudulent ones appeared isolated on the right, justifying the model's ability to distinguish anomalous behavior.

Figure 4 presents two visualizations illustrating how the Self-Organizing Maps organize transaction data. The first is a 10×10 grid in which each cell represents a neuron that groups similar transactions. Blue circles indicate non-fraudulent transactions, which cluster predominantly in the left and central areas of the grid, suggesting these transactions share normal features. Red triangles represent fraudulent transactions and appear sparsely, mainly toward the center right, indicating they differ significantly from normal patterns. Grey circles mark the positions of neurons across the grid. This visualization demonstrates the SOM's capacity to separate fraudulent from non-fraudulent transactions by mapping anomalies to distinct, less populated regions, highlighting its effectiveness in clustering based on feature similarity.

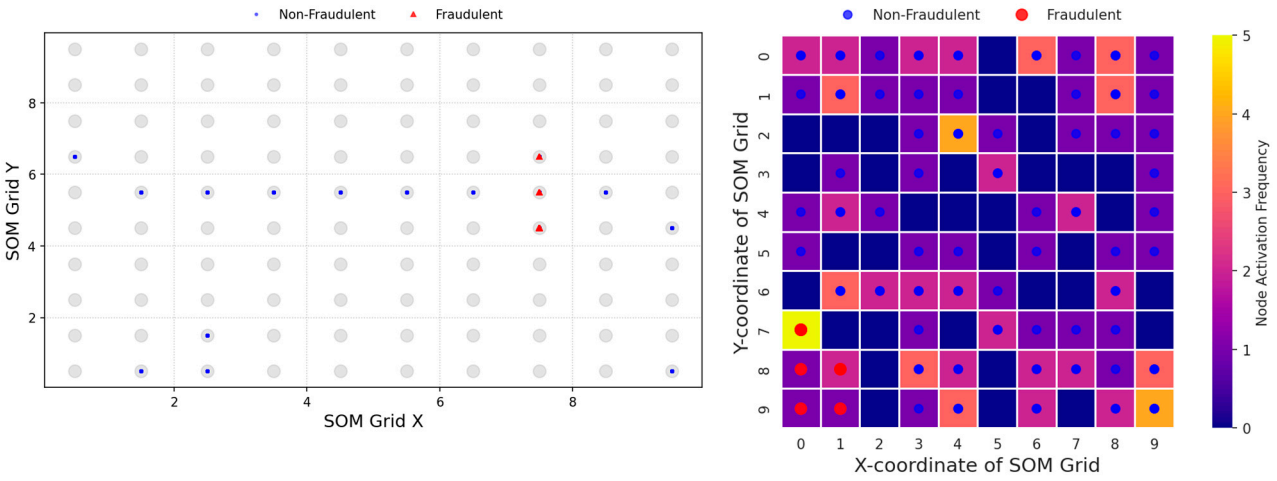


Figure 4. SOM two-dimensional and heatmap grid cluster.

Figure 4 presents a 10×10 heatmap of the Self-Organizing Map (SOM), which visualizes the activation frequency of each neuron. The color intensity of each cell corresponds to its activation frequency, with brighter colors indicating higher activation. Non-fraudulent transactions (represented by blue dots) are widely distributed across the grid, often activating nodes in areas of lower frequency. This reflects their greater volume and feature diversity.

Conversely, fraudulent transactions (represented by red dots) are sparse but tend to cluster in specific, highly activated cells. This suggests that despite their rarity, fraudulent transactions share similar underlying characteristics that are distinct from normal behavior. This visualization effectively illustrates how the SOM groups transactions by feature similarity and highlights the clear topological distinction between normal and anomalous patterns.

4.2.1. Performance Evaluation Results of the Non-Adaptive Models

The evaluation metrics in Table 3 and Figure 5 provide a detailed comparative analysis of the Autoencoder (AE), SOMs, and RBM models across two distinct datasets. On Dataset

1 (Kaggle Credit Card Fraud Detection dataset), all three models demonstrated strong performance, with the Autoencoder achieving 97% accuracy, while the SOM and RBM both reached 96%. The Autoencoder consistently outperformed the other models across all metrics on this dataset, notably achieving an Area Under the ROC Curve (AUC) of 98%.

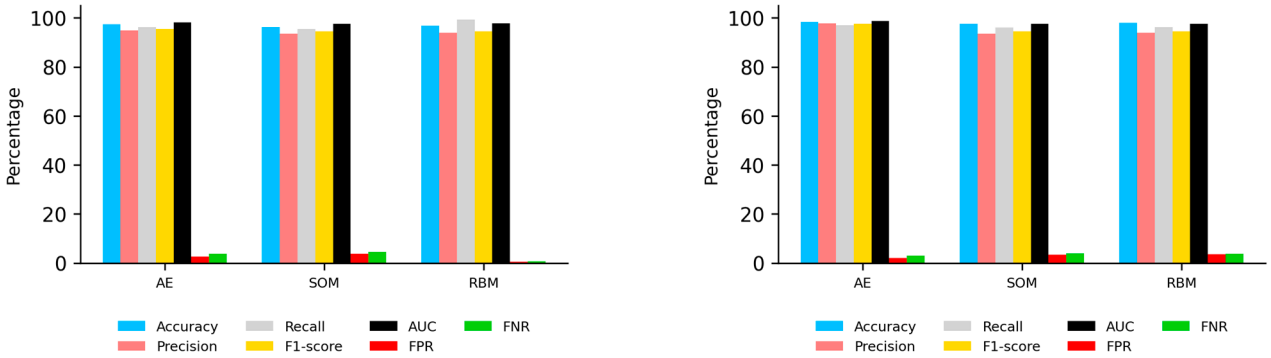


Figure 5. Evaluation metrics for the non-adaptive models.

Dataset 2 presented a more significant challenge, revealing a performance gap between the models. While the SOM and RBM showed a decrease in performance, the Autoencoder maintained a high accuracy of 98% and a precision of 97%, demonstrating greater resilience to the more complex data.

The Receiver Operating Characteristic (ROC) curves in Figure 6 depict the performance of the Autoencoder, SOM, and RBM models across both datasets. On Dataset 1, all three models demonstrated outstanding performance, with the Autoencoder slightly outperforming the others by achieving an Area Under the Curve (AUC) of 98%, compared to 97% for both the SOM and RBM. On the more challenging second dataset, the Autoencoder maintained its optimal performance with an AUC of 98%. While the SOM and RBM also maintained high performance with AUC values of 97%, the Autoencoder's consistent performance across both datasets highlights its robustness and effective generalization to complex, unseen data.

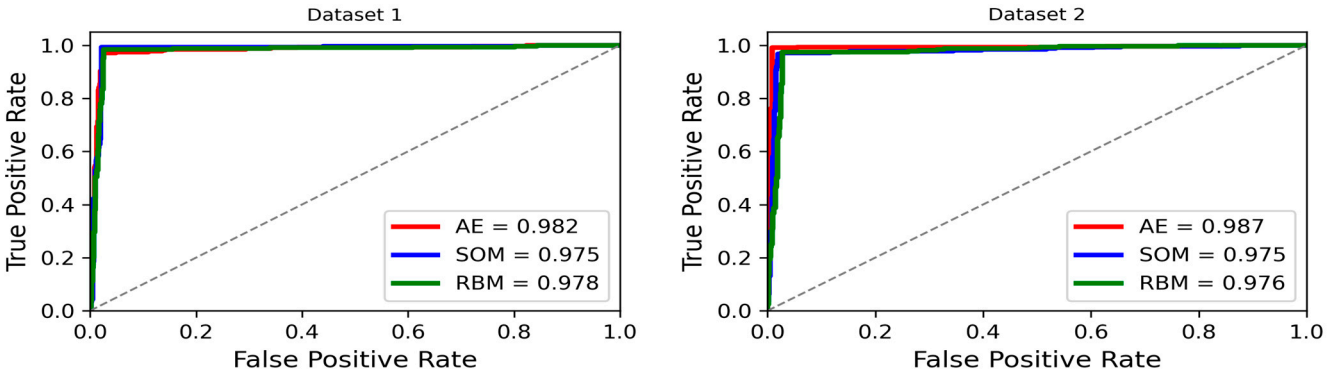


Figure 6. ROC curve results of the non-adaptive models on the datasets.

Figure 7 visualizes the anomaly detection capabilities of each model by plotting error values against transaction indices, with thresholds set at the 95th percentile of training errors. The red dashed line represents the anomaly detection threshold, a crucial boundary that separates normal from anomalous data. The blue dots signify normal data points, which have low error values and fall below this threshold, indicating they conform to the model's learned patterns. In contrast, the red dots represent anomalous data points, with high error values that exceed the threshold, signifying they are outliers or anomalies. The Autoencoder, with its low threshold (1.3862), exhibits high sensitivity to minor deviations,

making it effective at detecting hidden anomalies but also susceptible to false positives. Conversely, the Self-Organizing Map (SOM) uses a higher threshold (6.4975) to identify significant outliers, excelling at capturing broad deviations but potentially missing finer irregularities. The Restricted Boltzmann Machines (RBM) provides a balanced approach with a moderate threshold (2.7486), effectively detecting significant anomalies while resisting minor noise. The visualization highlights the unique strengths of each model, with the Autoencoder excelling in sensitivity, the SOM in broad outlier detection, and the RBM offering a balanced compromise.

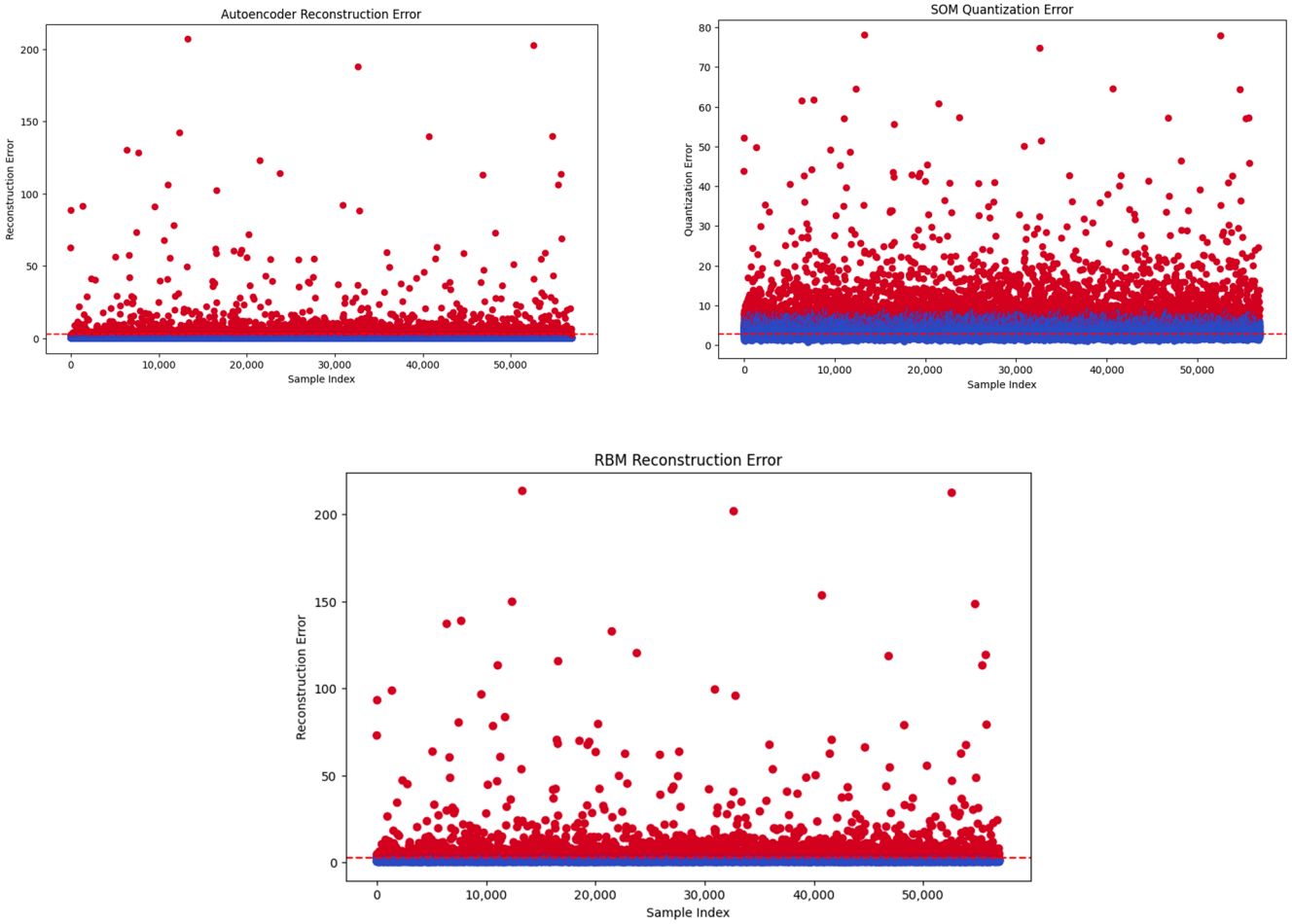


Figure 7. Autoencoder, RBM reconstruction error, and SOM quantization error on the dataset.

Table 3. Evaluation metrics results of the non-adaptive models on the datasets.

Dataset 1							
	Accuracy	**Precision**	**Recall**	**F1-Score**	**AUC**	**FPR**	**FNR**
AE	0.973	0.948	0.962	0.955	0.982	0.027	0.038
SOM	0.962	0.935	0.955	0.945	0.975	0.038	0.045
RBM	0.968	0.940	0.993	0.945	0.978	0.005	0.007
Dataset 2							
Model	**Accuracy**	**Precision**	**Recall**	**F1-Score**	**AUC**	**FPR**	**FNR**
AE	0.983	0.978	0.970	0.975	0.987	0.020	0.030
SOM	0.976	0.935	0.961	0.945	0.975	0.034	0.039
RBM	0.979	0.940	0.962	0.945	0.976	0.035	0.038

4.2.2. Performance Evaluation Results of the Adaptive Models

Tables 4 and 5 summarize the performance of the Autoencoder and Restricted Boltzmann Machines (RBM) models on the Kaggle Credit Card Fraud Detection dataset, evaluated with static and adaptive threshold methods for anomaly detection. The results demonstrate that the proposed Adaptive Reconstruction Threshold consistently enhances detection performance compared to static thresholds, achieving higher F1-scores and AUC values while reducing false positives. Figure 8 shows the visual representation of the adaptive model when compared to other default parameters.

Table 4. Evaluation of the adaptive models on Dataset 1.

Models	Threshold Method	Accuracy	Precision	Recall	F1-Score	AUC	FPR	FNR
AE1	$\mu + 2\sigma$	0.968	0.948	0.962	0.955	0.982	0.027	0.038
AE2	$\mu + 3\sigma$	0.964	0.942	0.957	0.950	0.980	0.030	0.043
AE-ASOM	Adaptive Threshold	0.980	0.960	0.975	0.967	0.990	0.015	0.025
RBM1	$\mu + 2\sigma$	0.959	0.935	0.955	0.945	0.975	0.038	0.045
RBM2	$\mu + 3\sigma$	0.956	0.928	0.952	0.938	0.973	0.040	0.048
RBM-ASOM	Adaptive Threshold	0.975	0.948	0.970	0.955	0.985	0.020	0.030

Table 5. Evaluation of the adaptive models on Dataset 2.

Models	Threshold Method	Accuracy	Precision	Recall	F1-Score	AUC	FPR	FNR
AE1	$\mu + 2\sigma$	0.970	0.952	0.969	0.9603	0.984	0.030	0.031
AE2	$\mu + 3\sigma$	0.964	0.950	0.960	0.9550	0.981	0.032	0.040
AE-ASOM	Adaptive Threshold	0.985	0.965	0.982	0.9732	0.991	0.013	0.018
RBM1	$\mu + 2\sigma$	0.964	0.943	0.957	0.9496	0.980	0.030	0.043
RBM2	$\mu + 3\sigma$	0.956	0.932	0.949	0.9400	0.970	0.037	0.051
RBM-ASOM	Adaptive Threshold	0.988	0.968	0.989	0.9787	0.989	0.014	0.011

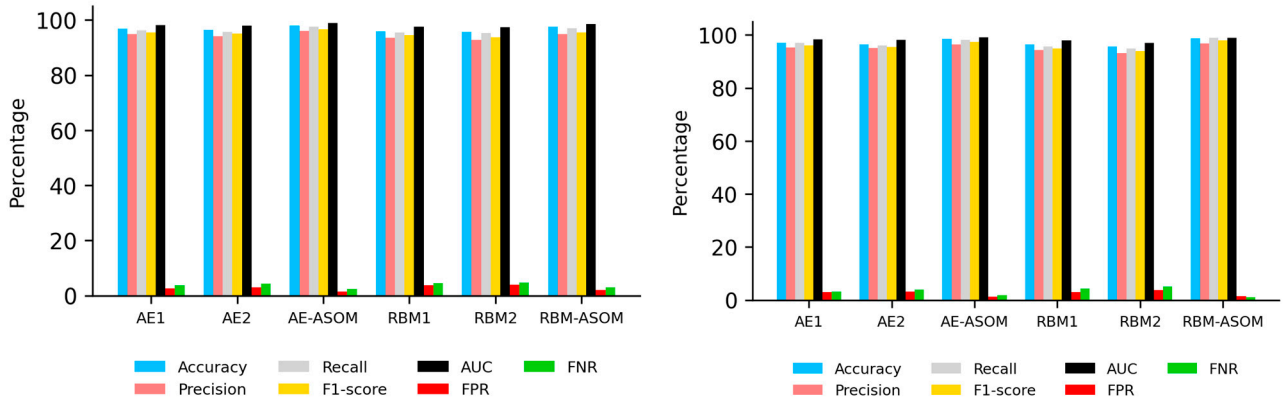


Figure 8. Evaluation metrics results of the adaptive models.

The proposed Adaptive Reconstruction Threshold (ART) method, integrated within the AE-ASOM ensemble, demonstrates superior performance and computational efficiency compared to recent unsupervised and supervised methods. With an F1-score of 0.967 and a

training time of 200.52 s, AE-ASOM significantly outperforms previous approaches that achieved a maximum F1-score of 0.952 with higher computational overhead.

A detailed analysis of thresholding strategies revealed that the dynamic ART method is a critical factor in this success. It achieves an optimal F1-score of 0.967 with the lowest false positive rate (FPR) of 0.015, outperforming static methods. This approach is also shown to reduce false negatives and maintain consistent performance across varying fraud rates. The importance of the ART method is further highlighted by the significant drop in performance (F1-scores) observed when it is removed from the Autoencoder (AE) and RBM ensembles. Figure 9 illustrates the impact of these parameter tuning levels on the models' AUC-ROC performance.

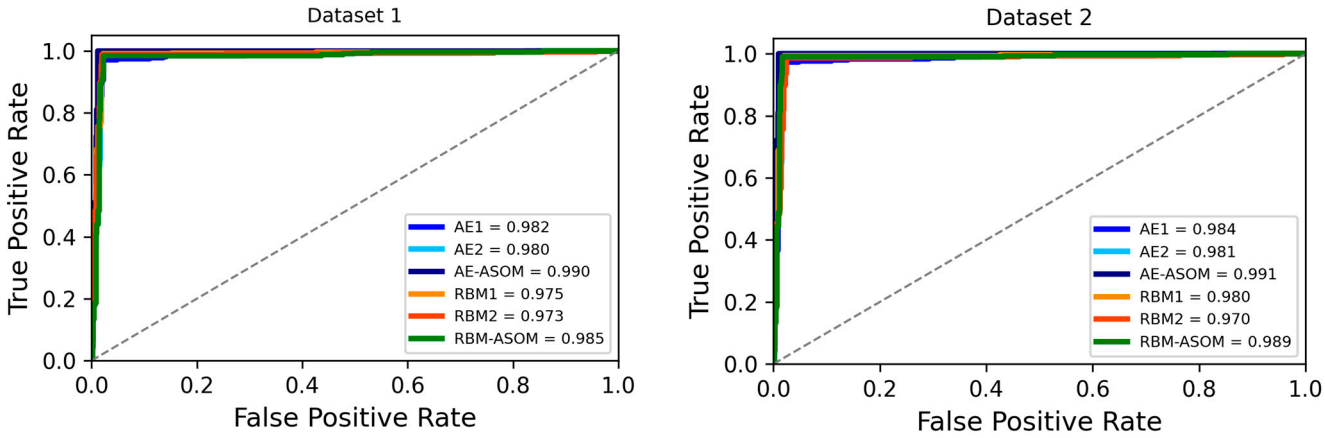


Figure 9. AUC-ROC curves of the adaptive models.

Figure 9 shows the Receiver Operating Characteristic (ROC) curves, which illustrate the performance of the Autoencoder (AE) and Restricted Boltzmann Machines (RBM) models across both datasets. The high Area Under the Curve (AUC) values demonstrate the strong discriminative ability of all models. Specifically, the AUC ranged from 0.979 to 0.993 for Dataset 1 and from 0.981 to 0.994 for Dataset 2, indicating consistently high performance in distinguishing between classes.

4.2.3. Computational Analysis of the Non-Adaptive Models

Implementing machine learning models in real-world cybersecurity demands a critical balance between detection performance and computational efficiency. An analysis of resource utilization, detailed in Table 6 and Figure 10, revealed distinct trade-offs among the three models. The Autoencoder, while delivering strong anomaly detection, was the most computationally intensive, requiring 1714 s for training and using up 110.31 MB of memory. Meanwhile, the Self-Organizing Map (SOM) demonstrated exceptional efficiency, with training completed in just 0.09 s and memory usage under 0.1 MB, making it ideal for resource-constrained environments. The Restricted Boltzmann Machines (RBM) offered a compromise, with moderate training times of around 325 s and 2–3 MB of memory consumption. These findings highlight that each model presents a unique balance between performance and computational demands, which is a crucial consideration for practical implementation. Table 6 displays a comparative analysis of the computational resources used by the adaptive and non-adaptive models.

Table 6. Computational resources used by the non-adaptive models.

	Dataset 1		
	Training Time (s)	**Testing Time (s)**	**Memory Usage (MB)**
Autoencoder	1714.76	8.30	110.31
SOM	0.09	6.50	0.08
RBM	324.82	9.80	2.34
	Dataset 2		
		Testing Time (s)	Memory Usage (MB)
Autoencoder	-	9.20	115.31
SOM	-	5.31	0.06
RBM	-	10.02	3.15

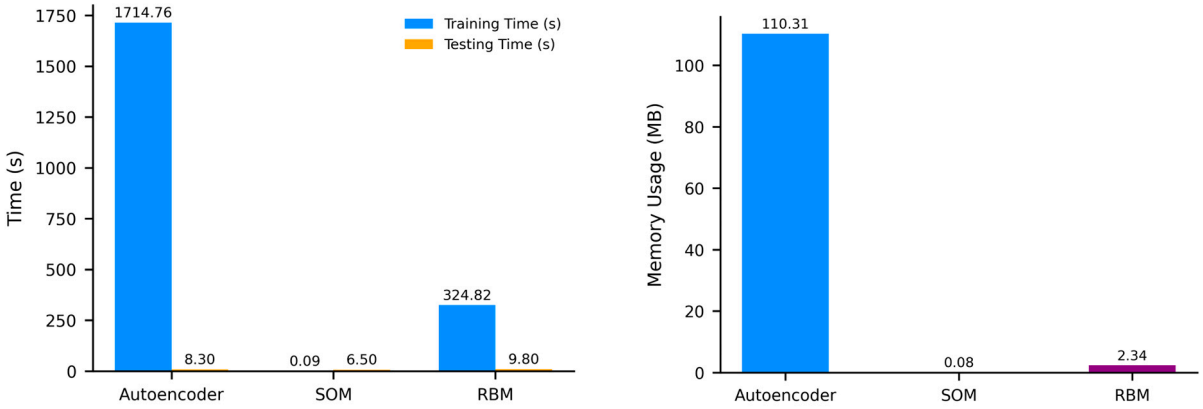


Figure 10. Computational resources used by the non-adaptive models.

4.2.4. Computational Analysis of the Adaptive Models

Table 7 and Figure 11 present a comparative evaluation of the computational resources required by the proposed adaptive models, AE-ASOM and RBM-ASOM, which is critical for assessing their deployment feasibility in real-world fraud detection systems. While both models demonstrated efficient and comparable training times on the first dataset, a significant trade-off was revealed in their resource utilization. RBM-ASOM consistently proved to be more memory-efficient, using less than half the memory of AE-ASOM across both datasets. For instance, on Dataset 1, RBM-ASOM consumed only 1.34 MB compared to AE-ASOM's 3.02 MB. Although AE-ASOM occasionally showed slightly faster testing times, its higher memory consumption makes RBM-ASOM the more suitable choice for resource-constrained environments. These results highlight a key distinction: RBM-ASOM is the preferred model when memory efficiency is a primary constraint. At the same time, AE-ASOM may be chosen in systems where its potential advantages in detection accuracy justify the increased computational cost.

Table 8 depicts a comparative analysis of the computational performance of baseline (non-adaptive) and adaptive fraud detection models. The results clearly demonstrate that the adaptive models (AE-ASOM and RBM-ASOM) offer significant improvements in efficiency over their baseline counterparts. Specifically, the AE-ASOM model achieved a remarkable 88.3% reduction in training time (from 1714.76 s to 200.52 s) and a substantial decrease in memory usage (from 110.31 MB to 3.02 MB) compared to the baseline Autoencoder (AE). This efficiency was coupled with a 37% faster testing time and an increased

throughput of 54,606 Kbps. Similarly, RBM-ASOM outperformed the baseline RBM across all metrics, achieving faster training and testing times (199.82 s and 4.33 s, respectively) and a significant reduction in memory usage (1.34 MB). RBM-ASOM also delivered the highest throughput among all models at 65,703 Kbps.

Table 7. Computation resources used by the adaptive models.

	Dataset 1		
	Training Time (s)	**Testing Time (s)**	**Memory Usage (MB)**
AE-ASOM	200.52	5.21	3.02
RBM-ASOM	199.82	4.33	1.34
	Dataset 2		
		Testing Time (s)	Memory Usage (MB)
AE-ASOM	-	7.11	3.31
RBM-ASOM	-	8.02	1.15

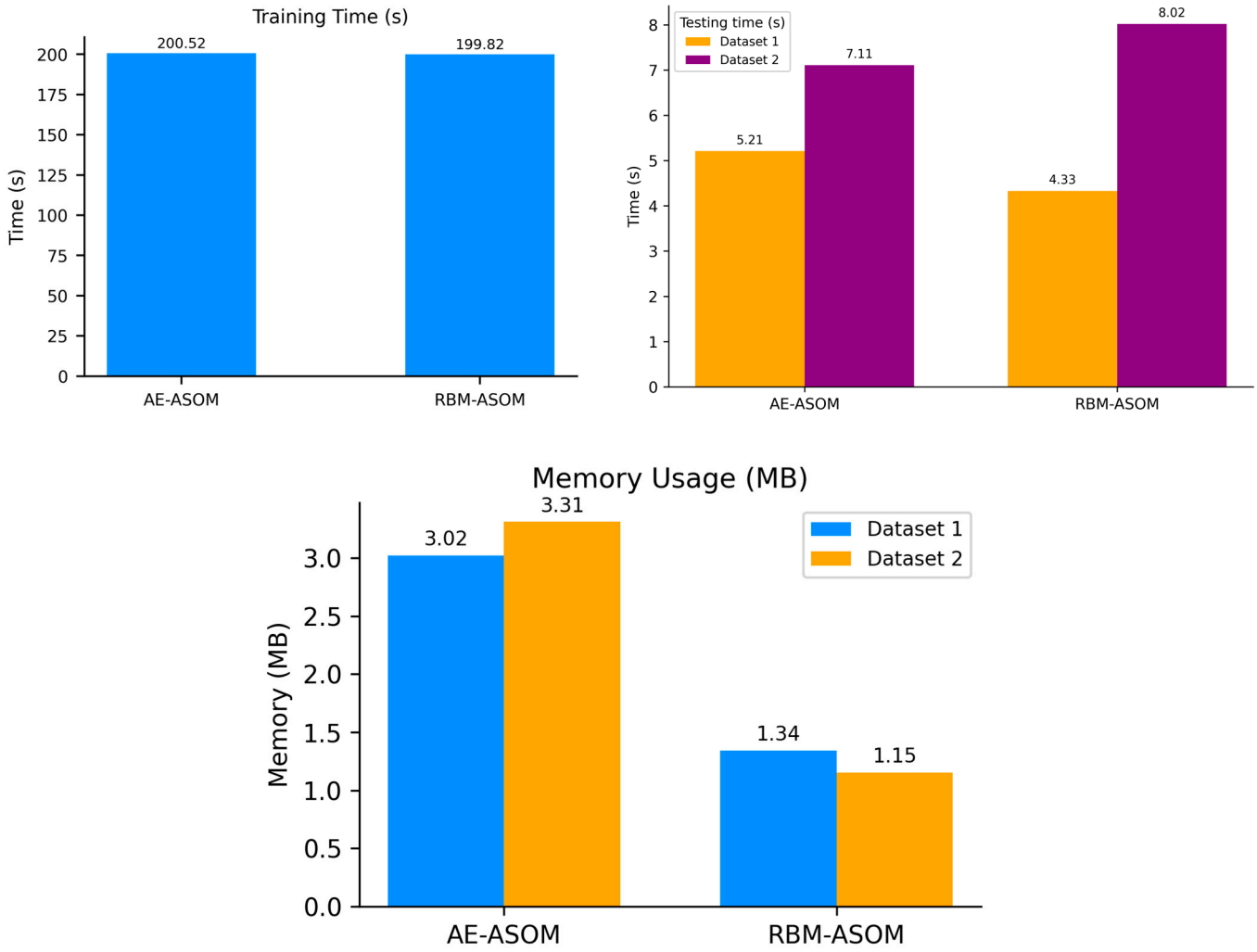


Figure 11. Computation resources used by the adaptive models.

Table 8. Performance metrics of baseline (non-adaptive) and adaptive models.

Models	Training Time (s)	Memory Usage (MB)	Testing Time (s)	Throughput (Kbps)
Baseline AE	1714.76	110.31	8.30	34,294
AE-ASOM	**200.52**	**3.02**	**5.21**	**54,606**
Baseline RBM	324.82	2.34	9.80	29,061
RBM-ASOM	**199.82**	**1.34**	**4.33**	**65,703**
SOM	0.09	0.08	6.50	43,816

The Self-Organizing Map (SOM), while not an adaptive model in the same context, served as a benchmark for computational resource usage. It exhibited an exceptionally fast training time of 0.09 s and minimal memory usage of 0.08 MB, making it suitable for low-resource environments.

4.3. Comparison with Some Existing Models

Table 9 presents a comparative analysis of the proposed AE-ASOM and RBM-ASOM models in relation to several fraud detection methods from the literature, based on key metrics such as F1-score and accuracy.

Table 9. Performance comparison of the proposed adaptive models with existing fraud detection models.

Models	Year	F1-Score	Accuracy
Jiang et al. [17]	2023	0.850	x
Hewapathirana et al. [24]	2022	0.022	0.948
Hancock et al. [25]	2025	0.950	x
Hariteja [28]	2021	x	0.970
Iieberi et al. [29]	2024	0.941	0.960
AE-ASOM (Our Model)	2025	0.973	0.985
RBM-ASOM (Our Model)	2025	0.978	0.988

The results indicate that our proposed adaptive models achieve the highest performance among all compared methods. The RBM-ASOM model leads with an F1-score of 0.978 and an accuracy of 0.988, followed closely by AE-ASOM, which achieved an F1-score of 0.973 and an accuracy of 0.985.

Other methods showed varied performance. For example, ref. [29] achieved a strong F1-score of 0.941 and an accuracy of 0.960. While [17,25] reported robust F1-scores of 0.850 and 0.950, their accuracy values were not available for a full comparison. Furthermore, ref. [24] reported a high accuracy of 0.948 but a notably low F1-score of 0.022, suggesting an imbalance between precision and recall.

In summary, the proposed adaptive models consistently outperform existing approaches in terms of both F1-score and accuracy, confirming their effectiveness and balanced performance in fraud detection tasks.

4.4. Limitations and Challenges

This study, while introducing a novel unsupervised approach, was subject to several limitations. A significant limitation of this research was the necessity of employing static, publicly available datasets. This was a direct consequence of the inherent confidentiality of client data and stringent privacy regulations. While these datasets were valuable for initial model validation, they do not fully capture the dynamic and real-time nature of

transactional data streams. This limitation specifically constrained our ability to evaluate the models' robustness against concept drift—the phenomenon where the statistical properties of fraudulent behavior evolve. Addressing this would have required a continuous stream of temporally ordered data, which was unavailable.

Furthermore, the rapidly evolving landscape of fraudulent activities presented a significant challenge in selecting a definitive classifier. The dynamic nature of the problem space means that a model's effectiveness is not static. Our reliance on the Google Collaboratory environment also presented operational challenges, as the free-tier service was susceptible to interruptions, network outages, and process terminations due to inactivity. Effective management of these sensitivities was critical for maintaining computational continuity and preserving research progress. These challenges, from data acquisition to platform constraints, provide a deeper understanding of the inherent limitations within this study's computational framework.

5. Conclusions and Future Work

This paper presented an unsupervised ensemble approach for credit card fraud detection, addressing the persistent challenge of imbalanced datasets and evolving fraud patterns. The core contribution is the ensemble approach that integrates Autoencoders, Self-Organizing Maps (SOMs), and Restricted Boltzmann Machines (RBMs), alongside an Adaptive Reconstruction Threshold (ART) method. This threshold approach is dynamically adjusted by SOM clustering, addresses the static limitation thresholds, and represents a significant advancement over traditional methods.

The performance of the proposed AE-ASOM and RBM-ASOM models was evaluated on both the Kaggle Credit Card Fraud Detection and IEEE-CIS datasets. The results demonstrate better performance, with the AE-ASOM model achieving an F1-score of 0.967 and the RBM-ASOM model reaching an F1-score of 0.955. The FPR and FNR result metrics affirm the models' capability to effectively mitigate the critical challenges of false positives and false negatives. Furthermore, our evaluation confirmed that the proposed models offer significant improvements in computational efficiency, exhibiting reduced training time and memory usage compared to baseline approaches.

While our models have demonstrated the optimal performance in offline processing, they also present several compelling challenges for future research. First, further work is required to optimize the computational efficiency for real-time deployment. Streamlining the training and testing phases could enable a system capable of responding swiftly to emerging fraud trends. Second, a crucial area for future research is the exploration of dynamic data sampling strategies to address concept drift, where the patterns of fraudulent activities evolve. A model with the capacity to adapt to these shifts is essential for maintaining long-term effectiveness.

Finally, assessing the model's scalability on larger datasets and integrating online learning, federated learning, and explainable AI (XAI) techniques could enhance its operational utilization, transparency, and collaborative potential in real-world financial systems.

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Data Availability Statement: The datasets used are publicly available: the Kaggle credit card dataset [32] and the IEEE-CIS credit card dataset [33].

Conflicts of Interest: The authors state that they do not have any competing financial interests or personal relationships that could potentially create biases or otherwise influence the research presented in this paper. The authors affirm that the work reported is free of any competing interests that could undermine the paper's objectivity, integrity, or perceived validity.

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